Examiner only

5. A combined heat and power station uses the wasted heat energy from a power station to provide heating for local homes and businesses. This increases the efficiency of the energy transfers.

(a) Describe how electricity is produced in a fossil fuel power station. [3]

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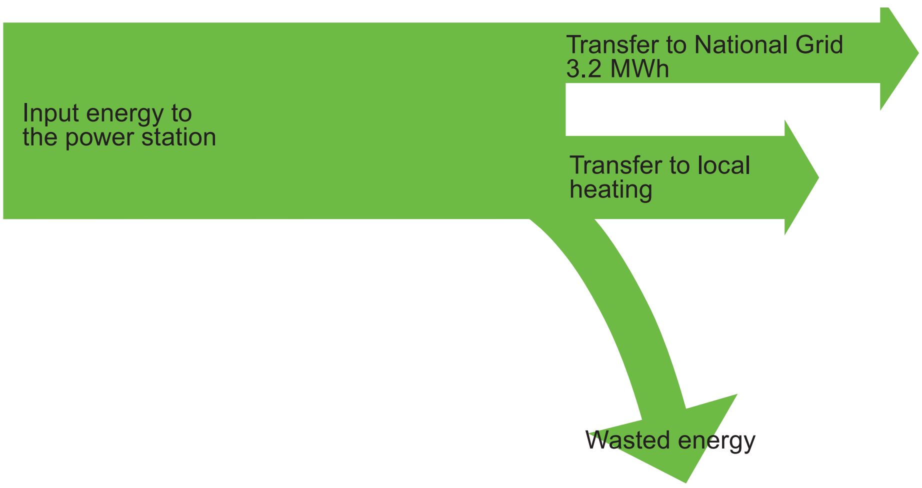
(b) The Sankey diagram shows the energy transfers for a combined heat and power station in a housing development. 75 % of the input energy to the power station is transferred as useful energy, some to the National Grid and some to local heating. 40 % of the useful energy produced by the power station is transferred to the National Grid. Helen estimates the input energy to the power station to be 20 MWh. Determine if this estimate is correct. [3]



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(c) Another power station produces 1 200 MW at a voltage of 50 kV. This voltage is increased to 400 kV by a step-up transformer before being transmitted by the National Grid. Use an equation from page 2 to calculate the current in the wires of the grid. Assume that the transformer is 100% efficient. [3]

Current = ... A

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